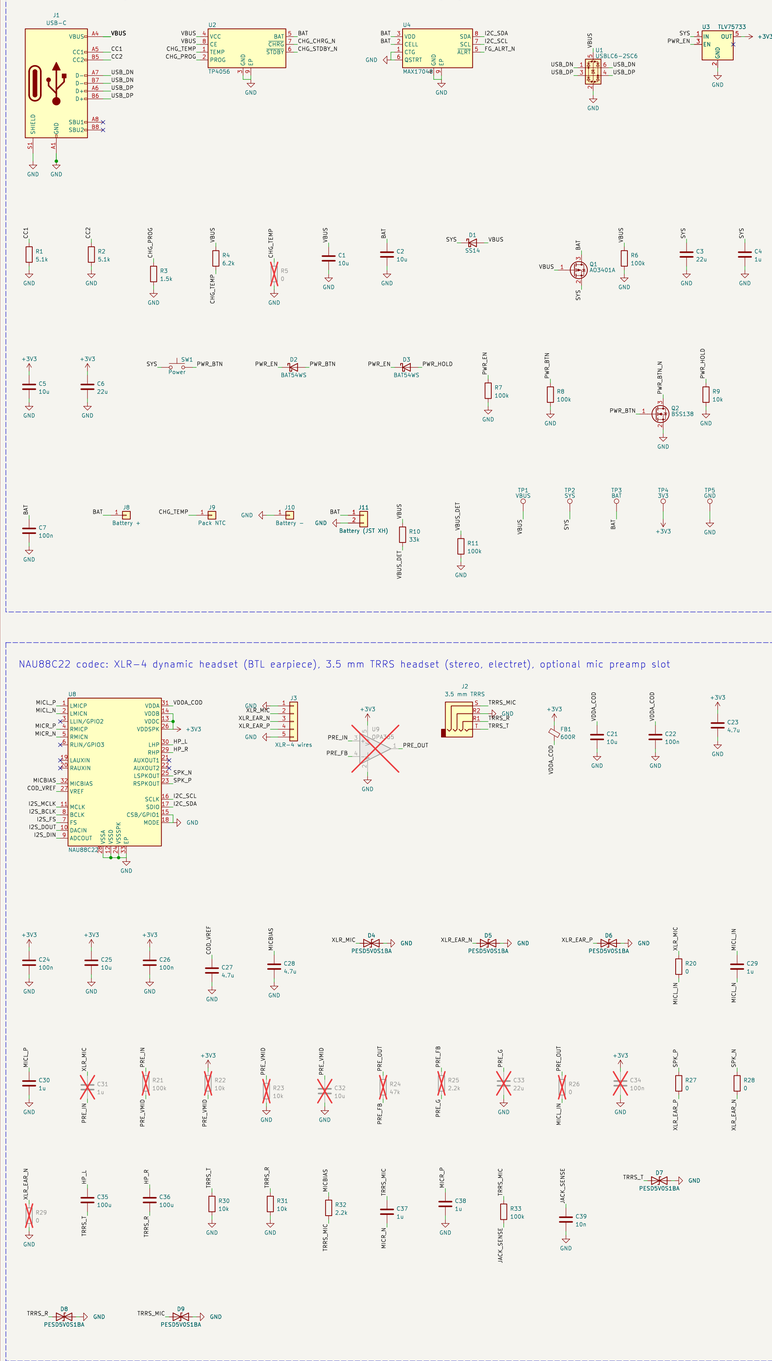
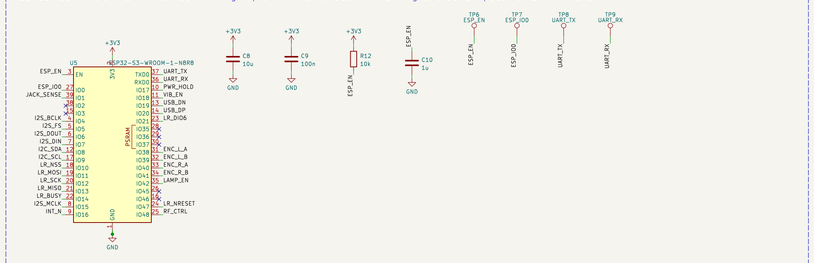
OpenTheatreComms beltpack — main PCB

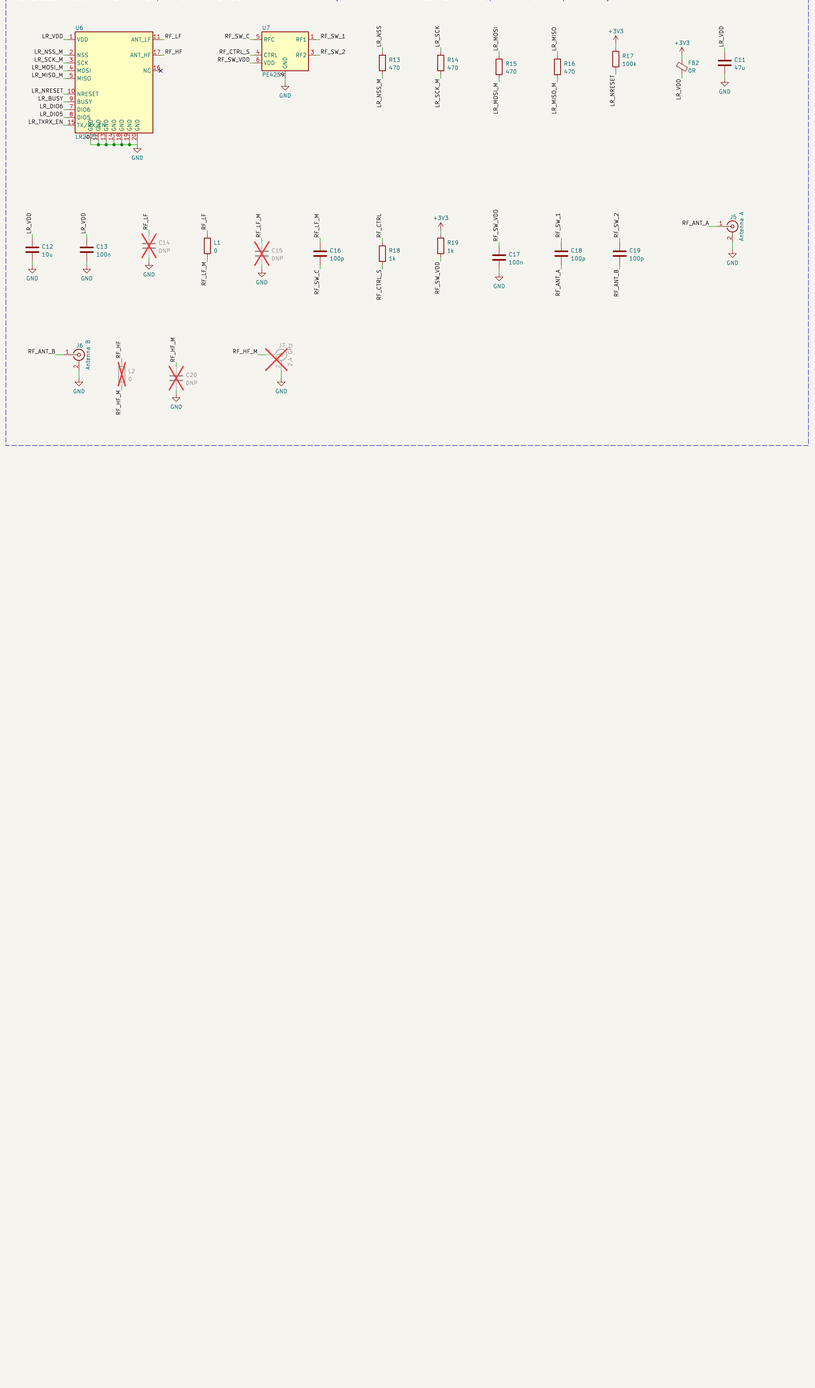
Power: USB-C, charger with power path, 3.3 V LDO with soft power switch, fuel gauge, battery pins



ESP32-S3-WROOM-1-N8R8: radio timing, Opus, UI. Native USB for flashing and serial; pads for EN, IO0 and UART0

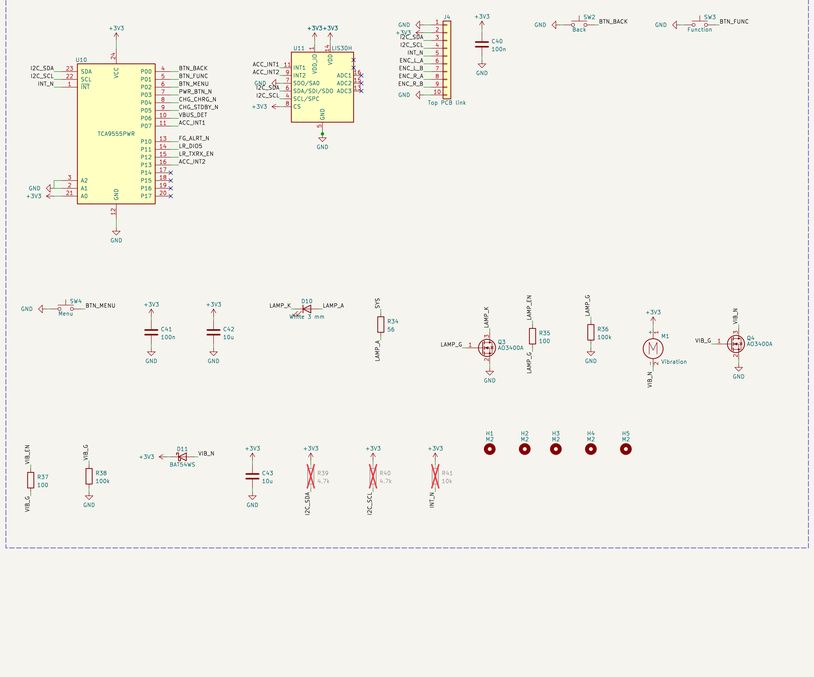


LR2021 module, 915 MHz pi match, PE4259 antenna diversity switch, two U.FLs; 2.4 GHz port as a footprint only



NAU88C22 codec: XLR-4 dynamic headset (BTL earpiece), 3.5 mm TRRS headset (stereo, electret), optional mic preamp slot

IO expander (0x21), front buttons, LIS3DH accelerometer, flashlight, vibration motor, link to the top PCB, mounting holes



Power, ESP32-S3, LR2021 radio and RF, codec and headsets, IO. Generated from pcb_main/design.py

OpenTheatreComms

Sheet: /

File: main_pcb.kicad_sch

Title: Beltpack main PCB

Size: A1	Date: 2026-10-26	**Rev: 0.1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1